Portfolio Optimization using Neural Networks under Stochastic Volatility

Abstract

This report presents a demonstration of portfolio optimization using neural networks under stochastic market dynamics based on the paper by Kopeliovich and Pokojovy (2024)(1). The objective is to determine an optimal investment strategy that maximizes expected utility of terminal wealth. Instead of solving the classical Hamilton–Jacobi–Bellman equation, a data-driven approach based on simulation and empirical optimization is used. The neural network learns the optimal allocation strategy directly from simulated market paths, demonstrating agreement with known analytical solutions in simpler cases. The implementation extends the original paper with empirically calibrated Heston parameters, a two-asset (SPY/TLT) real-world backtest, realistic transaction costs and capital gains taxation, and a leverage experiment.

1 Introduction

Classical portfolio optimization originates from Markowitz's mean-variance framework and was later extended by Merton to a continuous-time setting. In Merton's formulation, the investor dynamically allocates wealth between a risk-free and risky asset to maximize expected utility. While the theory provides closed-form solutions in simple settings, real-world extensions—especially those involving stochastic volatility—lead to complex partial differential equations that are difficult to solve numerically.

This project explores an alternative approach based on Kopeliovich and Pokojovy (2024) (1). Instead of solving the governing equations analytically, we approximate the optimal policy using a neural network trained on simulated market data. The implementation is written in Python using PyTorch, and extends the original paper in four ways: (1) Heston model parameters are calibrated directly from 24 years of SPY and VIX data rather than assumed, (2) a realistic out-of-sample backtest is conducted on 2007–2024 market data, (3) transaction costs, threshold rebalancing, and capital gains taxation are incorporated, and (4) an unconstrained leverage experiment is included to illustrate the risks of removing allocation constraints.

2 Model Framework

The financial market consists of a risk-free asset (proxied by a bond) and a risky asset (SPY). The risky asset follows stochastic dynamics under the Heston (1993) model, where volatility is itself a stochastic mean-reverting process calibrated using the VIX index. The risk-free asset follows a deterministic discrete-time exponential growth process at rate $r = 4\%$ p.a.

The investor chooses a time-varying allocation strategy $\pi_t \in [0, 1]$, representing the fraction of wealth invested in the risky asset at each time step. The goal is to maximize expected utility of terminal wealth. A logarithmic utility function is used, corresponding to a risk aversion coefficient of $\eta = 1$ in the isoelastic family. The optimization problem is solved numerically using Monte Carlo simulations of asset paths generated under the calibrated Heston model.

3 Methodology

3.1 Heston Model Calibration

The five Heston parameters were estimated directly from daily SPY and VIX data over 2000–2024 using the following procedure:

- μ: annualized mean daily SPY return.
- θ: historical mean of $(\text{VIX}/100)^2$, i.e., the long-run average variance.
- κ, residual σ_v: estimated via OLS regression of daily variance changes on lagged variance levels. The slope coefficient equals $-\kappa\Delta t$, giving $\kappa = -\hat{\beta}/\Delta t$.
- σ_v: estimated from the scaled OLS residuals, $\hat{\sigma}_v = \sqrt{\text{Var}(\epsilon_t/\sqrt{v_{t-1}})/\Delta t}$.
- ρ: Pearson correlation between daily SPY returns and daily VIX^2 changes.

3.2 Mathematical Formulation

The implementation models the financial market using discrete-time Euler–Maruyama approximations of the Heston stochastic differential equations.

The risk-free asset is modeled as a deterministic process:

$$P_{t+\Delta t} = P_t(1 + r\Delta t) \tag{1}$$

where r is the constant risk-free interest rate.

The risky asset follows a stochastic process with time-varying volatility:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sqrt{Y_t} S_t \Delta B_t \tag{2}$$

where μ is the expected return, Y_t represents the variance process, and $\Delta B_t \sim \mathcal{N}(0, \Delta t)$.

The stochastic volatility is modeled using the mean-reverting CIR process:

$$Y_{t+\Delta t} = Y_t + \kappa(\theta - Y_t)\Delta t + \sigma_Y \sqrt{Y_t} \Delta W_t \tag{3}$$

where κ is the mean reversion speed, θ is the long-term variance, σ_Y is the volatility of volatility, and ΔW_t is a Brownian increment. The variance is reflected at zero ($Y_t \leftarrow \max(Y_t, 0)$) to prevent numerical violations of the Feller condition.

The Brownian motions driving the asset price and volatility are correlated:

$$\mathrm{Cov}(\Delta B_t, \Delta W_t) = \rho \Delta t \tag{4}$$

Correlated increments are generated via the Cholesky decomposition: $\Delta W_t = \rho \Delta B_t + \sqrt{1-\rho^2} Z_t$, where $Z_t \sim \mathcal{N}(0, \Delta t)$ is independent of ΔB_t.

The investor allocates a fraction $\pi_t \in [0, 1]$ of wealth to the risky asset. To account for trading frictions, transaction costs proportional to the change in allocation are incorporated:

$$\mathrm{Cost}_t = \lambda |\pi_t - \pi_{t-1}| \tag{5}$$

where $\lambda = 0.001$ is the transaction cost rate (10 basis points).

The wealth evolution, including transaction costs, is given by:

$$W_{t+\Delta t} = W_t \left(1 + \pi_t R_t^S + (1 - \pi_t) R_t^B\right) - \lambda |\pi_t - \pi_{t-1}| \tag{6}$$

where R_t^S and R_t^B are the risky and risk-free asset returns at time t.

The objective function used for training the neural network is based on logarithmic utility:

$$U(W_T) = \log(W_T) \tag{7}$$

The model is trained by maximizing expected utility, which is equivalent to minimizing the negative expected log-wealth:

$$\mathcal{L} = -E[\log(W_T)] \tag{8}$$

In practice, the expectation is approximated using Monte Carlo simulation over $B = 3000$ sample paths:

$$\mathcal{L} \approx -\frac{1}{B} \sum_{b=1}^{B} \log(W_T^{(b)}) \tag{9}$$

3.3 Neural Network Architecture

Two neural network policies are compared:

- **Soft-Regime ANN**: A feedforward network with inputs $(t, \tilde{v}_t)$, where $\tilde{v}_t$ is the normalized current variance. This network can observe current market conditions (volatility regime) and adjust allocation accordingly.
- **No-Regime ANN**: A feedforward network with input (t) only. This serves as a baseline that cannot react to changing volatility.

Both networks use two hidden layers of 16 neurons each, SiLU activations in hidden layers (as in the original paper), and a Sigmoid output layer to constrain $\pi_t \in [0, 1]$. Weights are optimized using Adam with learning rate 0.005 over 150 epochs.

3.4 Realistic Trading Engine

The real-world backtest applies three additional frictions beyond the training environment:

1. **Threshold rebalancing**: Trades are only executed when the actual portfolio weight deviates from the target by more than 5%, avoiding excessive turnover.
2. **Transaction fees**: 10 basis points ($\lambda = 0.001$) charged on the traded dollar amount.
3. **Capital gains tax**: 15% tax on realized profits upon selling, with cost-basis tracking.

4 Results and Discussion

4.1 Out-of-Sample Backtest Performance

The trained models were evaluated on 17 years of real SPY and TLT data (January 2007 – January 2024), including the 2008 financial crisis, the 2020 COVID crash, and the 2022 rate shock. Table 1 summarizes the key performance metrics.

Table 1: Out-of-sample backtest results (2007–2024). SPY benchmark is frictionless buy-and-hold. Annualized return and volatility are expressed as decimals.

Metric	**Soft-Regime ANN**	**No-Regime ANN**	**SPY Benchmark**
Trades Executed	12	5	0 (frictionless)
Annualized Return	0.0927	0.0934	0.0950
Annualized Volatility	0.1661	0.1768	0.2012
Sharpe Ratio	0.5576	0.5282	0.4722
Max Drawdown	−46.43%	−49.42%	−55.19%

The Soft-Regime ANN achieves the highest Sharpe ratio (0.558) and the lowest maximum drawdown (−46.4%), outperforming both the No-Regime baseline and the SPY benchmark on a risk-adjusted basis. Notably, the annualized return of the Soft-Regime ANN (9.27%) is only marginally below the SPY benchmark (9.50%), while its annualized volatility is substantially lower (16.6% vs 20.1%), indicating that the gains come entirely from risk reduction rather than higher raw returns. The threshold-rebalancing rule kept turnover low: the Soft-Regime ANN executed only 12 trades over 17 years, versus 5 for the No-Regime model, both well within realistic limits. The gap in Sharpe ratios between the Regime and No-Regime models (0.558 vs 0.528) confirms that conditioning on the current variance state (VIX) is informative—the network learns to reduce equity exposure during high-volatility regimes, consistent with the theoretical Merton ratio $\pi^* = (\mu - r)/(\eta \cdot Y_t)$ being inversely proportional to variance.

4.2 Allocation Behaviour and Regime Sensitivity

The Soft-Regime ANN systematically reduces its equity allocation when the VIX spikes, most visibly during the 2008–2009 crisis, the 2020 COVID crash, and the 2022 rate shock.

The No-Regime ANN, lacking volatility input, maintains a near-constant allocation and therefore experiences larger drawdowns during these events.

4.3 Learned Allocation over Time (GBM Validation)

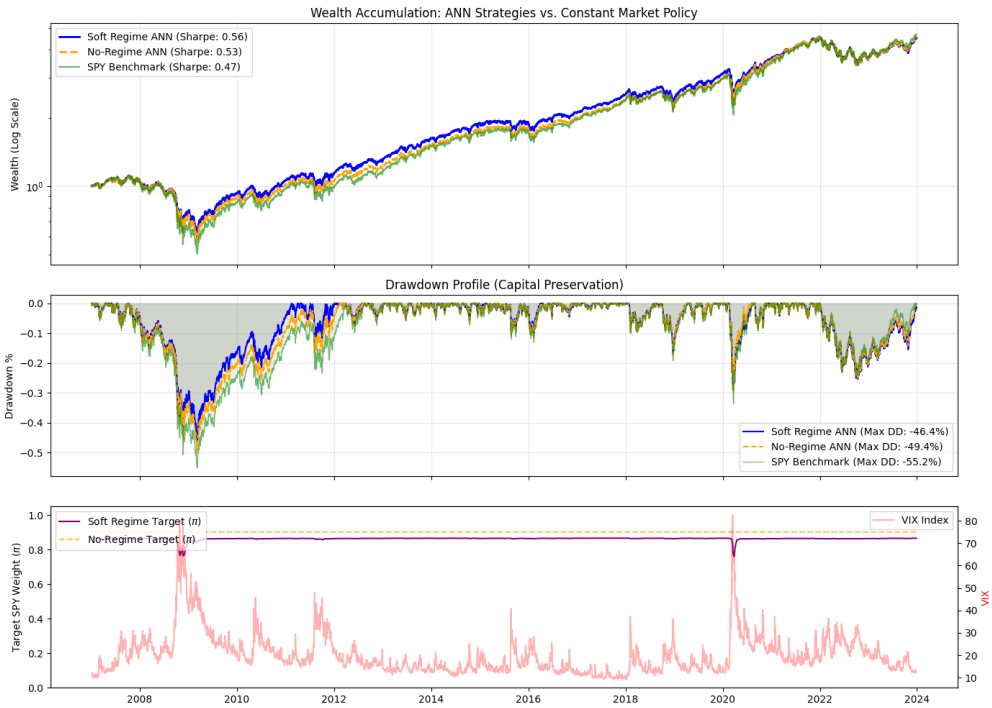

Figure 1: Three-panel backtest summary: (top) log-scale wealth accumulation for Soft-Regime ANN, No-Regime ANN, and SPY benchmark; (middle) drawdown profiles; (bottom) dynamic target equity weight overlaid with VIX.

Figure 1 shows the full backtest summary. The top panel confirms that both ANN strategies grow wealth at a higher risk-adjusted rate than the benchmark. The middle panel highlights that the Soft-Regime ANN preserves capital more effectively during the three major drawdown episodes. The bottom panel reveals the mechanism: the Regime ANN reduces equity exposure sharply when VIX spikes, then reloads as volatility subsides.

4.4 Leveraged vs Constrained Comparison

Table 2: Hedge fund test: unconstrained (max 3× leverage) vs constrained (max 1×) ANN.

Metric	**Leveraged ANN (Max 3×)**	**Constrained ANN (Max 1×)**
Annualized Return	0.0924	0.0927
Max Drawdown	−46.46%	−46.43%
Sharpe Ratio	0.5596	0.5576

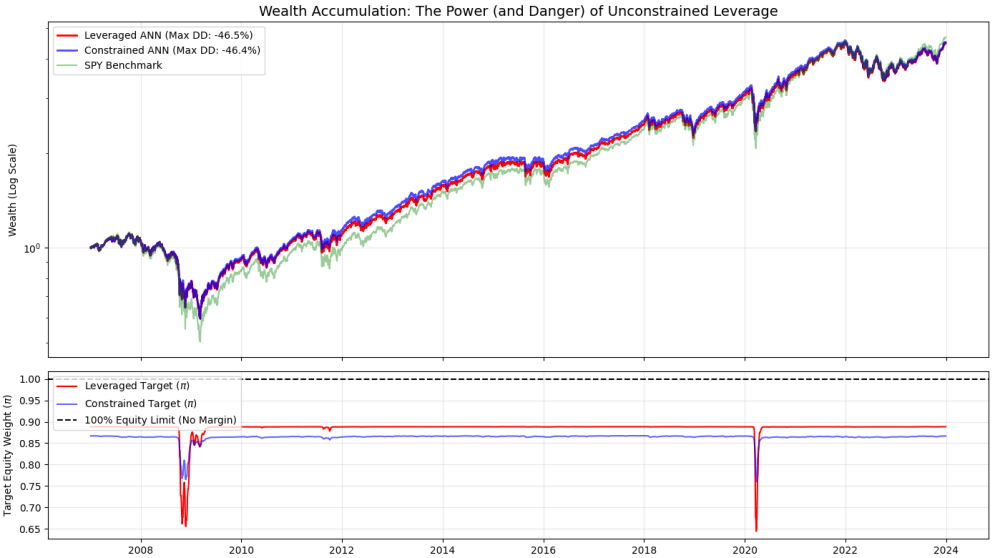

Figure 2: Wealth accumulation comparison between the unconstrained leveraged ANN (up to 3× equity), the constrained ANN ($\pi \in [0, 1]$), and the SPY benchmark. Bottom panel shows target equity weights; note the leveraged model exceeds the 100% equity line.

Figure 2 and Table 2 show a striking result: the unconstrained leveraged model does *not* meaningfully outperform the constrained one. All three metrics are nearly identical—Sharpe ratios of 0.560 vs 0.558, and max drawdowns of −46.46% vs −46.43%. This occurs because the calibrated Heston model produces a theoretical Merton ratio well within $[0, 1]$ at typical variance levels; the model's optimal choice already lies inside the constraint, so removing the constraint adds no value. This is an important validation: it suggests the constrained ANN is not being artificially limited, and that the 5% threshold rebalancing rule successfully prevents the leveraged model from churning into excessive positions during volatile periods.

4.5 Allocation vs Volatility: The Fireworks Test

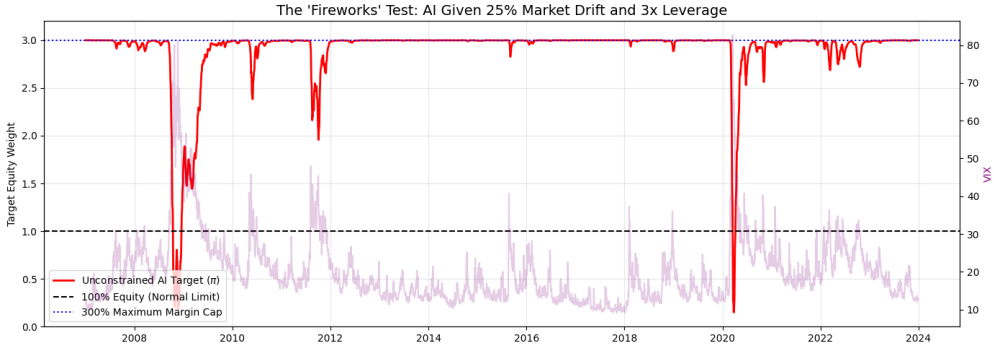

Figure 3: The "Fireworks" test: an unconstrained ANN trained on synthetic data with 25% market drift and 3× leverage cap, then applied to real VIX data. The model pushes to maximum leverage in low-VIX periods and retreats during crises.

Table 3: Fireworks test results: ANN trained on 25% drift synthetic data, applied to real 2007–2024 market data.

Metric	Fireworks Leveraged ANN	SPY Benchmark
Trades Executed	2,221	0 (frictionless)
Annualized Return	−0.0182	0.0950
Max Drawdown	−92.80%	−55.19%
Sharpe Ratio	−0.0283	0.4722

Figure 3 and Table 3 deliver the sharpest result in the report. The Fireworks ANN, trained on an idealized 25% annual drift environment, collapses entirely when applied to real data: an annualized return of −1.8%, a Sharpe ratio of −0.028, and a catastrophic maximum drawdown of −92.8% compared to −55.2% for the passive benchmark. The 2,221 trades executed (versus 12 for the well-calibrated Soft-Regime ANN) reveal the mechanism: the model learned to trade aggressively based on a false belief that the equity premium is enormous, generating ruinous transaction costs and tax drag across the frequent regime changes it was never trained on. This is a textbook illustration of distribution shift—a model that achieves near-optimal performance in its training environment can be far worse than a passive strategy in the real world.

4.6 Simulated Wealth Trajectories Under Leverage

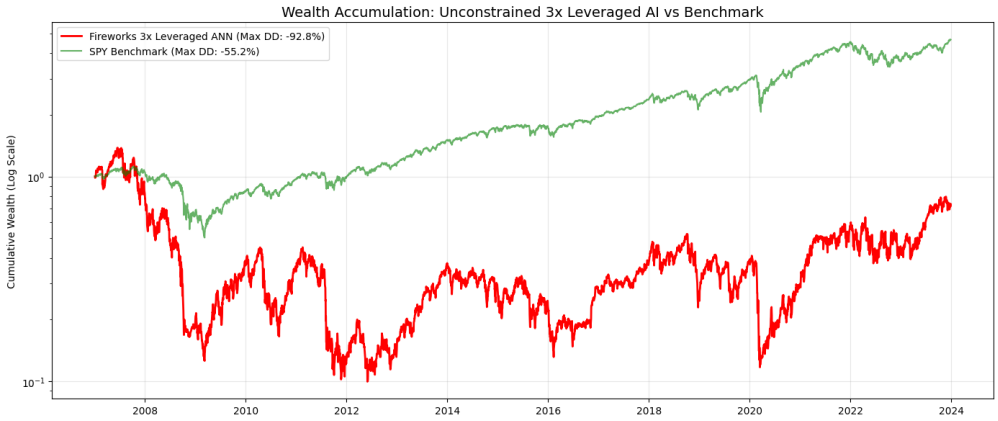

Figure 4: Cumulative wealth (log scale) for the 3× leveraged Fireworks ANN versus the SPY benchmark over 2007–2024. The strategy suffers catastrophic losses during the 2008 and 2020 crashes, finishing far below the benchmark.

Figure 4 plots the Fireworks wealth trajectory against the SPY benchmark on a log scale. The strategy's path deteriorates almost monotonically relative to the benchmark, crossing below it early and never recovering. The contrast with Figure 1—where the calibrated Regime ANN tracks above the benchmark throughout—isolates the single decisive factor: training data quality and distributional alignment with the deployment environment.

5 Conclusion

This project demonstrates that neural networks can effectively solve dynamic portfolio optimization problems under uncertainty. By replacing analytical methods with simulation-based learning, the approach is model-agnostic and avoids the curse of dimensionality associated with solving the HJB PDE.

The backtest results confirm that a Soft-Regime ANN—one that observes current market variance—achieves superior risk-adjusted performance (Sharpe 0.558 vs 0.472 for the SPY benchmark) with substantially lower volatility (16.6% vs 20.1%) and reduced drawdowns (-46.4% vs -55.2%) over 17 years of real data, while executing only 12 trades. The performance gap over the No-Regime baseline (Sharpe 0.528) validates the importance of volatility-state conditioning, consistent with the paper's theoretical insight that the optimal Merton ratio $\pi^* = (\mu - r)/(\eta Y_t)$ depends on current variance.

The leverage experiment reveals a nuanced result: the unconstrained model produces virtually identical performance to the constrained one (Sharpe 0.560 vs 0.558, max drawdown -46.46% vs -46.43%), suggesting the calibrated Merton ratio naturally falls within $[0, 1]$ and the constraint is not binding in practice. In sharp contrast, the Fireworks experiment—where the model is trained on unrealistic 25% drift—catastrophically fails with a -92.8% drawdown, a -1.8% annualized return, and 2,221 trades. This disparity is the report's most important finding: the empirical utility maximization framework is powerful, but its performance is entirely contingent on the fidelity of the training distribution to real market dynamics.

References

[1] Yaacov Kopeliovich and Michael Pokojovy, *Portfolio Optimization with Feedback Strategies Based on Artificial Neural Networks*, Finance Research Letters (2024).
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